Section 3: Stablecoin Balance Sheets, the Merton Model, and Risk Management Under the GENIUS Act

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Abstract:

This section applies the Merton structural credit-risk framework to analyze a payment stablecoin issuer under the first comprehensive federal regulatory framework for dollar-denominated stablecoins, the Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS) Act of 2025. This report formalizes the issuer's balance sheet in the viewpoint of the Act's 1:1 reserve requirement, derives the default threshold and the equity-as-call-option result and explains why the incentive geometry of a GENIUS-compliant issuer is quite distinct from that of a standard levered corporate borrower. We then examine two risk categories that are not captured directly by the Merton model: interest rate risk arising from a short-duration asset / zero-duration liability mismatch, and liquidity risk on technically solvent but operationally illiquid reserves and suggest practical and statute-compliant ways to address these risks. The analysis is based on the actual text of the Act and its 2026 implementing rulemakings (OCC, FDIC and Treasury), not just a theoretical exercise.

Index Terms: stablecoins, Merton model, GENIUS Act, bank runs, balance sheet, interest rate risk, liquidity risk, structural credit risk, payment systems

I. Introduction

According to the Merton model [1] a leveraged firm's equity can be thought of as a call option on the assets of the firm, exercised at the face value of the debt. The beauty of the model is that it reduces the analysis of solvency to three basic components: the market value of assets, the face value of liabilities and the volatility of the assets' value over the period of interest. This section applies that framework to a payment stablecoin issuer regulated under the GENIUS Act, signed into law on 18 July 2025 [4] which mandates that every payment stablecoin be backed no less than 1:1 by a very narrowly drawn group of cash and short-term U.S. Treasury instruments [5]. The reserve, disclosure, and redemption provisions of the Act are not arbitrary regulatory ornamentation, but, as Section III demonstrates, they closely correspond to the volatility and default-threshold terms in the Merton formula, and their absence is exactly what destabilized earlier, less-regulated stablecoin designs.

The discussion is divided into four sections which correspond to the four parts of the Assessment Brief: balance sheet construction and the Merton default threshold (Section II); issuer incentives and equity as a call option (Section III); interest rate risk from the required collateral structure of the Act (Section IV); and liquidity risk and redemption-run dynamics (Section V).

II. The Merton Model and Balance Sheet Construction

A. The GENIUS-Compliant Issuer's Balance Sheet

A permitted payment stablecoin issuer (PPSI) under the GENIUS Act may hold reserves only in: (i) U.S. coins, currency, and balances at a Federal Reserve Bank; (ii) demand deposits at insured depository institutions; (iii) Treasury bills, notes or bonds with a remaining maturity of 93 days or less; (iv) overnight repurchase agreements collateralised by such Treasury bills; (v) short-dated reverse repurchase agreements similarly collateralised; and (vi) shares in money market funds invested solely in the foregoing [5], [6]. Reserves cannot be pledged, rehypothecated or used for other purposes except under very limited conditions (when providing margining on an allowed repo, custodial servicing or to satisfy liquidity requirements

for redemptions) [5]. Importantly, issuers are not allowed to pay interest or yield to stablecoin holders [7] and are not subject to the bank regulatory capital regime applicable to insured depository institutions [8].

This produces a balance sheet that is structurally simpler and economically stranger than a conventional bank's:

$$A_t = L_t + E_t \qquad (1)$$

where A_t is the market value of the reserve assets (cash, deposits and short-term Treasury instruments) at time t; L_t is the face value of the stablecoins issued, which is by construction equal to the number of outstanding stablecoins times the $1.00 par redemption value, because the stablecoin issuer is contractually obligated to redeem at par; and E_t is the residual claim, or issuer equity.

There are two characteristics that make this different from a textbook Merton firm that need to be articulated prior to using the model:

1) The liability is not a zero-coupon bond maturing at a fixed time T, but it is a perpetual, instantly demandable claim. A stablecoin holder can redeem at any time as opposed to only at a contractually agreed upon maturity. This is more like a demand deposit or a money-market-fund share than a corporate debt and therefore it is the 'first passage' version of the model that should be used (default is possible as soon as A_t falls below L_t, not only at a terminal date) even though the simpler terminal-default intuition is enough to establish the core results below.

2) The 1:1 reserve mandate compresses E_0 towards zero by statutory design, rather than leaving leverage as a discretionary managerial decision. The capital structure is decided by the conventional Merton firm, but the GENIUS-compliant issuer is required to always hold $A_t \geq L_t$, which implies $E_t \geq 0$, as a regulatory requirement (in addition to the requirement as a no-arbitrage residual of the model) which is enforced through the reserve disclosures mandated by the Act on a monthly basis, and third-party attestations [5].

B. The Default Threshold

In the Merton sense, default occurs when reserve assets are insufficient to cover redemption claims of stablecoin holders:

$$\text{Default at } t \iff A_t < L_t \qquad (2)$$

Because L_t is fixed in nominal terms (each token is a claim on exactly \$1.00) while A_t fluctuates with the mark-to-market value of the underlying Treasury instruments (formalised in Section IV) and with any credit or operational losses on deposits or repo counterparties, the "distance to default" for a stablecoin issuer can be expressed analogously to the standard Merton distance-to-default statistic:

$$DD_t = (A_t - L_t) / (\sigma_a \sqrt{(T - t)}) \qquad (3)$$

where σ_a represents the volatility of the value of the reserves. The whole prudential apparatus of the GENIUS Act could be interpreted as an effort to make the numerator of Eq. (3) as close to a known, audited, non-negative constant as possible (through the 1:1 mandate and monthly attestation) and to force σ_a to the lower bound it is possible to achieve for any interest-bearing asset (through the 93-day maturity ceiling on eligible Treasury instruments). By design, a GENIUS-compliant issuer should have a large distance to default, where the denominator behaviour under interest rate stress (Section IV) is the dominant factor, rather than the credit risk of the assets, and that is why the later sections of this assessment, dealing with interest rate and liquidity risk, rather than credit risk, are the relevant threats to a well-run, compliant issuer.

III. Issuer Incentives and Equity as a Call Option

A. Equity as a Call Option on Reserve Assets

The key point of the Merton model is that the value of the issuer equity can be thought of as a European call option on the value of the reserve assets with a strike price of the face value of the redemption liability:

$$E_T = \max(A_T - L_T, 0) \qquad (4)$$

Equation (4) is the standard Merton payoff, and Black-Scholes option-pricing logic applies directly: E_t is increasing in σ_a (asset volatility), all else equal. This is the foundation of the well-documented risk-shifting or asset-substitution incentive [2] that the GENIUS Act is, in large part, legislatively designed to prevent.

B. Why 'Reaching for Yield' Is the Central Incentive Problem Here

In a traditional levered firm, the incentive for the equity holders to increase σ_a is seen in the way they prefer to invest in riskier projects that are financed using debt from creditors who do not reprice the debt completely for added risk. For a stablecoin issuer, the analogous temptation is direct and unusually clean. The issuer gets all the difference between the rate of return on the assets it holds as reserves and the (statutorily required) rate of return that it pays to stablecoin holders, which is zero. Since issuers cannot pass on any yield to holders [7], each basis point of additional yield extracted from the reserve portfolio accrues entirely to issuer equity, and the list of eligible assets in the GENIUS Act is just the limit on how much additional yield (and hence how much additional σ_a) an issuer can chase. Within the permitted universe, the model still predicts a directional incentive, such that an issuer will rationally prefer the longest-permitted maturity (93 days, rather than overnight), and the higher-yielding instrument within each eligible category (e.g. repo over demand deposits, where repo carries incremental counterparty and rollover risk [12]), because both choices raise expected reserve income at some cost in σ_a. This is exactly the call-option-value trade implied by Eq. (4).

C. Regulatory Responses Implied by the Model

Three of the mechanisms of the GENIUS Act can best be seen as being direct constraints on the σ_a term (which in the Merton framework is the equity-holder incentive problem) rather than generic "consumer protection" mechanisms:

1) The 93-day maturity limit directly bounds interest-rate-driven mark-to-market volatility on the asset side (formalised in Section IV), which is the dominant component of σ_a once credit risk on Treasury instruments is assumed to be negligible.
2) The prohibition of rehypothecation (except in the narrow circumstances listed in Section II) prevents the issuer from synthetically increasing leverage on the asset side, which would otherwise allow an issuer to increase σ_a without overtly violating the 1:1 nominal reserve ratio.
3) Mandatory monthly reserve disclosure and third-party examination [5] shortens the information lag between an issuer drifting toward higher-risk asset choices and the market (and regulators) observing that drift. In the Merton framework, this is functionally equivalent to reducing the effective $T - t$ over which an undetected asset-substitution strategy could run before being priced into market confidence in the peg.

This is also why the necessity of strict regulatory frameworks to protect peg stability is not merely a policy platitude: it follows directly from the call-option geometry of Eq. (4), applied to an issuer with essentially zero required capital buffer (Section II) and a legislated, unidirectional incentive to maximise reserve-portfolio yield.

IV. Interest Rate Risk and Stablecoin Design

A. The Duration Mismatch

The stablecoin liability L_t has, by construction, a Macaulay duration of zero: it is a demandable, fixed-par claim with no scheduled cash flows and no price sensitivity to interest rates. A $1.00 stablecoin is worth $1.00 regardless of where Treasury yields sit. The reserve asset side, even constrained to instruments of 93 days or less, has strictly positive (if small) duration. An approximation for the price sensitivity of a portfolio of T-bills with average remaining maturity of $\bar{m}$, to a parallel shift Δy in yields is

$$\Delta A / A \approx -\bar{m} \cdot \Delta y \qquad (5)$$

Even with $\bar{m}$ capped at roughly a quarter-year, a sudden, sharp rate shock can move A_t below L_t at the margin, especially for an issuer already operating with a thin equity buffer (Section II). Such a shock would be analogous in direction, though much smaller in magnitude given the short maturity ceiling, to the duration mismatch that triggered the March 2023 failure of Silicon Valley Bank, whose long-duration held-to-maturity Treasury and agency-MBS portfolio lost substantial mark-to-market value as rates rose [15]. The 93-day limit in the GENIUS Act is explicitly a duration cap, and is the key statutory mechanism provided by the Act to manage this risk.

However, a cap on duration is not the same as eliminating duration risk; it merely bounds its magnitude.

B. Proposed Architecture for Interest Rate Risk Management

Building on the eligible-asset menu permitted under the Act [5], a prudently designed architecture of an issuer's reserves should include the following, each of which can be implemented directly in (that is without amendment to) the existing statutory framework:

1) Active duration laddering toward the short end of the permitted range: An issuer can choose to voluntarily ladder maturities towards overnight and one-to-seven-day instruments (repo, reverse repo and demand deposits), even though this might result in a lower expected yield (the incentive identified in Section III), but it has the effect of directly reducing the $\bar{m}$ term in Eq. (5). This is exactly the issue that the Merton/incentive analysis in Section III identifies issuers as having a private incentive to resist, and it is also the argument for a supervisory liquidity standard rather than reliance on voluntary issuer discipline.

2) A mandatory minimum-liquidity tier as required by a regulator, like the one that was proposed in the 2026 implementing rulemaking. The OCC's proposed three-level liquidity framework for federally chartered issuers includes holding at least 10% of outstanding stablecoins in cash or Federal Reserve balances that can be redeemed same day, at least 30% in high-quality liquid assets that can be redeemed within five business days and the remainder in standard reserve assets [9]. This tiering is functionally a regulator-imposed floor on the share of the portfolio held at (near-) zero duration, addressing exactly the duration mismatch problem formalized in Eq. (5) while still permitting some yield-bearing allocation in the higher tiers.

3) Floating-rate/overnight-indexed instruments where eligible. Within the GENIUS Act's permitted categories, overnight repo and reverse repo (already eligible per Section II) are the closest available floating-rate substitute for a fixed-rate 93-day bill: their value resets daily rather than being locked in at issuance, which eliminates the mark-to-market duration exposure of Eq. (5) almost entirely, at the cost of daily rollover/counterparty risk that must itself be managed (e.g. by using the tri-party or SEC-registered-clearing-agency structures already required for repo eligibility under the Act [5]).

4) A voluntary capital buffer above the statutory 1:1 minimum. Since the Act does not impose bank capital regulation on issuers [8], there is no externally imposed minimum equity buffer; an issuer that nonetheless retains a modest buffer (financed from its own yield-spread income, consistent with the incentive structure in Section III) directly widens the numerator of the distance-to-default statistic in Eq. (3), which would buffer the effects of the A_t decline from an interest-rate shock modelled in Eq. (5), without requiring any change to the statutory minimum.

V. Liquidity Risk and Run Dynamics

A. Why Solvency Is Not Sufficient

The Merton framework in Section II characterises solvency: whether $A_t \geq L_t$ in present-value, marked-to-market terms. It does not describe liquidity, namely, how fast and at what price A_t can be converted into cash, and how much the issuer can convert, so it can satisfy redemptions in case of a sudden, concentrated wave of redemption requests. This is the classic "Diamond–Dybvig" insight [3] that a balance sheet can be solvent at the prevailing market price of its assets and still fail if forced to liquidate those assets faster than the market can absorb the sale without a price concession, exactly the scenario described in the Treasury 2026 advance notice of proposed rulemaking under the GENIUS Act, which models a scenario in which an issuer forced to sell $10 billion of Treasury bills at even a 1% loss would realise a $100 million reserve shortfall. Absent an equity buffer, that shortfall would immediately breach the 1:1 requirement and could intensify rather than resolve the run [10].

B. Three Structural Features That Amplify (Rather Than Merely Permit) Stablecoin Run Risk

In addition to the generic Diamond–Dybvig mechanism, there are three aspects of the GENIUS Act's architecture that are mentioned in the policy literature after the Act was enacted as intensifying, not just enabling, the dynamics of runs and should be considered in any proposed mitigation:

1) No lender-of-last-resort access: Section 4(a)(8) of the GENIUS Act explicitly denies PPSIs access to the Federal Reserve's discount window, the Standing Repo Facility, and Section 13(3) emergency credit, and the Act separately clarifies that nothing in it expands issuers' eligibility for a Federal Reserve master account [11], [13]. A GENIUS-compliant issuer (as opposed to a commercial bank) cannot borrow at the "discount window" of the central bank in the event of a stablecoin run; it must fire-sell its asset portfolio, which means that it bears materially higher fire-sale risk for an economically identical liquidity shock.
2) Settlement-speed asymmetry: Stablecoins settle on blockchain rails operating continuously (24 hours a day, 7 days a week), and the market to which reserve assets need to be finally converted is the traditional Treasury market, which is only open during market hours [15]. A redemption wave which comes outside the Treasury market

hours cannot be met with asset liquidation at the same moment even where the reserves of the issuer are, on a marked basis, fully sufficient.

3) No deposit insurance: The lack of a government backstop means a holder of a stablecoin does not have to wait passively during a stress event. This removes the single mechanism (deposit insurance) credited with halting most historical retail bank runs once introduced [14], and this gives the rational stablecoin holder a clear incentive to redeem first and ask questions later as soon as the stablecoin issuer is distressed.

D. Proposed Liquidity Risk Mitigation Mechanisms

1) Tiered reserve structures, formalised rather than left to issuer discretion: As explained in Section IV, the OCC's proposed same day/five day/standard liquidity tiers [9] explicitly target liquidity risk, rather than just interest rate risk: a minimum requirement to keep a certain percentage of reserves in literal cash and Federal Reserve balances (the most liquid eligible asset class in Section II's list) provides a floor for the ability to redeem that is independent of same-day Treasury market liquidity.
2) Pre-arranged, rules-based redemption gates: Treasury's proposed regulations for 2026 explicitly consider the possibility of regulations that would automatically suspend new issuance and redemption when reserves are below the statutory level, rather than leaving the decision to an individual issuer’s discretion in the middle of a run [10]. A pre-committed trigger is in line with the broader literature on banking regulation, which argues that in the case of a bank run, it is better to have a pre-committed rather than an ex-post discretionary trigger as the latter would give the issuer an incentive to postpone an inevitable gate, hoping the bank run will pass, thereby exacerbating the eventual fire-sale.
3) Repo counterparty relationships which are diversified and pre-negotiated, with a registered clearing agency. The Act already requires that the repo be conducted via an SEC registered clearing agency or with the prior approval of the regulator [5] and so an issuer can pre-establish standing, clearable repo lines as a contingent liquidity source, enabling same-day cash generation against Treasury collateral without an outright asset sale into a stressed market. This arrangement is a direct, statute-compliant alternative to the discount-window access the Act denies (Section V.B), although of course the

same-day cash generation would still be subject to counterparty capacity constraints in the event of a systemic (as opposed to issuer-specific) stress.

4) Arbitrageur-network depth as a de facto liquidity buffer: As Brookings' comparative analysis of stablecoin redemption mechanics highlights, there are large differences in the number of active arbitrageurs willing to deal directly with the stablecoin issuer at or near par. Brookings reports a monthly average of roughly six active arbitrageurs for one major issuer versus several hundred for another [13]. A well-established and diversified network of arbitrageurs spreads the redemption pressure among many balance-sheet-light intermediaries instead of putting all the pressure on the issuer's own liquidity reserves on the balance sheet, and acts as a kind of market-structure complement to the issuer's own tiered reserves.
5) Access to central-bank facilities, contingent on future legislative or regulatory reform: While the current law explicitly denies PPSIs access to the discount window and to a master account, the academic and policy discussion since the law was enacted (for example, the Wharton/Federal Reserve discussion of whether issuers eventually should be given access to a Federal Reserve account [15]) suggests that this is a live, not settled, policy question. It is a longer-horizon mitigation path beyond the issuer's unilateral control, but worth stating explicitly as the brief invites “theoretical access to central bank facilities”. This would be a future, narrowly scoped GENIUS Act amendment granting large, well-supervised PPSIs contingent access to a Federal Reserve liquidity facility analogous to the discount window, though distinct from it in eligibility terms, in exchange for accepting bank-style supervisory and capital obligations currently exempted under the Act [8].

VI. Conclusion

The core mechanics of the Merton model, namely equity as a call option on reserve assets, default as the asset value crossing a fixed liability threshold, and the risk-shifting incentive that follows directly from the option's convexity in asset volatility, map onto a GENIUS Act-compliant stablecoin issuer with unusual clarity. The interpretation of the Act's rules on the composition of reserves, anti-rehypothecation and disclosure requirements as direct and deliberate limits on the asset-volatility term in the Merton framework, not as standalone consumer-protection rules, thus provides an explanation for the structure of the Act and a

clarification of where its protection genuinely ends: at the boundary between solvency (which the Act regulates tightly) and liquidity (which it leaves comparatively exposed, having denied issuers lender-of-last-resort access by explicit statutory design).

Finally, the interest rate and liquidity risk-management framework outlined in Sections IV–V is designed to operate fully within the Act's eligible-asset and permitted-activity boundaries, several elements of which (the OCC's proposed liquidity tiers, and Treasury's proposed automatic redemption-freeze trigger) are independently affirmed by 2026 implementing rulemaking. This convergence is itself supporting evidence that the analysis above identifies genuine, rather than merely hypothetical, vulnerabilities in the post-GENIUS stablecoin architecture.

Bibliography

[1] R. C. Merton, “On the Pricing of Corporate Debt: the Risk Structure of Interest Rates,” *The Journal of Finance*, vol. 29, no. 2, pp. 449–470, May 1974, doi: 10.2307/2978814.

[2] D. W. Diamond and P. H. Dybvig, “Bank Runs, Deposit Insurance, and Liquidity,” *Journal of Political Economy*, vol. 91, no. 3, pp. 401–419, 1983, doi: https://doi.org/10.1086/261155.

[3] M. Jensen and W. Meckling, *“Theory of the firm: Managerial behavior, Agency Costs and Ownership structure,” Corporate Governance*. Gower, 2000. doi: 10.4324/9781315191157.

[4] The White House, “Fact Sheet: President Donald J. Trump Signs GENIUS Act into Law,” *The White House*, July 18, 2025. https://www.whitehouse.gov/fact-sheets/2025/07/fact-sheet-president-donald-j-trump-signs-genius-act-into-law/ (accessed July 03, 2026).

[5] U.S. Congress, “S.1582 - 119th Congress (2025-2026): GENIUS Act,” *Congress.gov*, 2025. https://www.congress.gov/bill/119th-congress/senate-bill/1582/text (accessed July 03, 2026).

[6] R. Manfredi, “The GENIUS Act: a New Era of Stablecoin Regulation,” *Gibson Dunn*, July 18, 2025. https://www.gibsondunn.com/the-genius-act-a-new-era-of-stablecoin-regulation/ (accessed July 03, 2026).

[7] L. Ante, I. Fiedler, and E. Strehle, “The Influence of Stablecoin Issuances on Cryptocurrency Markets,” *Finance Research Letters*, vol. 41, p. 101867, Nov. 2020, doi: 10.1016/j.frl.2020.101867.

[8] U.S. Congress, “S.919 - 119th Congress (2025-2026): GENIUS Act of 2025,” *Congress.gov*, 2025. https://www.congress.gov/bill/119th-congress/senate-bill/919 (accessed July 03, 2026).

[9] R. Manfredi, “OCC Proposes Comprehensive Stablecoin Regulatory Framework to Implement the GENIUS Act,” *Gibson Dunn*, Mar. 03, 2026.

https://www.gibsondunn.com/occ-proposes-comprehensive-stablecoin-regulatory-framework-to-implement-the-genius-act/ (accessed July 03, 2026).

[10] P. Davies, “The Fall and Rise of Debt: Bank Capital Regulation after the Crisis,” *European Business Organization Law Review*, vol. 16, no. 3, pp. 491–512, Sept. 2015, doi: 10.1007/s40804-015-0017-8.

[11] Skadden, Arps, Slate, Meagher & Flom LLP, “US Establishes First Federal Regulatory Framework for Stablecoins: the GENIUS Act Passes Congress and Awaits President Trump’s Signature,” *Skadden.com*, July 17, 2025. https://www.skadden.com/insights/publications/2025/07/us-establishes-first-federal-regulatory-framework (accessed July 03, 2026).

[12] S. Hempel, R. Kahn, R. Mann, and M. Paddrik, “Brief Series Office of Financial Research,” 2024. Accessed: July 03, 2026. [Online]. Available: https://www.financialresearch.gov/briefs/files/OFRBrief-24-07-repo-market-intermediation.pdf

[13] N. Liang, “What Are the Differences between Payment Stablecoins and Tokenized Bank deposits?,” *Brookings*, Apr. 14, 2026. https://www.brookings.edu/articles/what-are-the-differences-between-payment-stablecoins-and-tokenized-bank-deposits/ (accessed July 03, 2026).

[14] S. Ugolini, *The Evolution of Central Banking: Theory and History*. London: Palgrave Macmillan UK, 2017. doi: 10.1057/978-1-137-48525-0.

[15] Wharton, “How Stablecoins Could Get More Stability with the GENIUS Act,” *Knowledge at Wharton*, Mar. 24, 2026. https://knowledge.wharton.upenn.edu/article/how-stablecoins-could-get-more-stability-with-the-genius-act/ (accessed July 03, 2026).